Optical Coherence Tomography High-Resolution Plaque Assessmen

Clinical Consensus Document · Part 25

Background and Rationale:

Optical Coherence Tomography (OCT) with 10-micron axial resolution. Differentiation of thin-cap fibroatheroma (TCFA), plaque erosion, macrophage infiltration, and stent strut malapposition.

Coronary artery disease involving complex anatomical subsets presents significant challenges. Technological advances including Intravascular Ultrasound (IVUS), Optical Coherence Tomography (OCT), Excimer Laser Coronary Atherectomy (ELCA), and Intravascular Lithotripsy (IVL) have revolutionized lesion preparation and stent implantation.

Indication Criteria for OCT:

1. Hemodynamic instability or ongoing ischemia with FFR < 0.80 or resting iFR < 0.89.
2. Severe calcification precluding stent delivery: recommended upfront treatment with IVL or ELCA.
3. In-Stent Restenosis (ISR): comprehensive assessment for under-expansion or neoatherosclerosis via OCT.
4. Left Main (LM) bifurcation disease requiring dedicated provisional or dual-stent strategy.

Section 2: Technical Specifications and Evidence Review

Detailed Procedural Steps (Protocol Step 1):

Radial access with 6F or 7F guide catheter is standard. Anticoagulation achieved with unfractionated heparin targeting an activated clotting time (ACT) of 250-300 seconds. Guide catheter extension (GuideLiner) is recommended for deep vessel support in tortuous calcified anatomy.

Stent sizing must adhere to distal reference vessel external elastic lamina (EEL) measured by IVUS. Post-dilation with high pressure non-compliant balloon (18-24 atm) is mandatory to prevent acute stent thrombosis.

Table 1.1 ? Nominal Pressures and Burst Pressures

Stent Diameter (mm)	Nominal (atm)	RBP (atm)	Target Minimum Area (mm²)
2.50 mm	9.0 atm	16.0 atm	4.5 mm²
3.00 mm	9.0 atm	18.0 atm	5.5 mm²
3.50 mm	10.0 atm	18.0 atm	7.0 mm²
4.00 mm (LM)	10.0 atm	20.0 atm	8.5 mm²

RBP: Rated Burst Pressure; LM: Left Main.